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Give the complexity of the following functions. Choose the most appropriate notation from among O , θ , and Ω .

```
1. void function1(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = i; j <= n; j += 2) {
            cout << "*";
        }
    }
}
```

Answer: $\theta(n^2)$

```
2. void function2(int n) {
    int count = 0;
    for (int i = 1; i * i * i <= n; i++) {
        count++;
    }
    cout << count;
}
```

Answer: $\theta(\sqrt[3]{n})$

```
3. void function3(int n) {
    int count = 0;
    for (int i = 1; i * i <= n; i++) {
        for (int j = 1; j + n/2 <= n; j++) {
            for (int k = 1; k <= n; k *= 2) {
                count++;
            }
        }
    }
    cout << count;
}
```

Answer: $\theta((n)^{3/2} * \log(n))$

```
4. void function4(int n) {
    int count = 0;
    for (int i = n/2; i <= n; i++) {
        for (int j = 1; j <= n; j *= 2) {
            for (int k = 1; k <= n; k *= 2) {
                count++;
            }
        }
    }
    cout << count;
}
```

Answer: $\theta(n * \log^2(n))$

```

5. void function5(int n) {
    if (n % 2 == 0) {
        return;
    }
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            cout << "*";
            break;
        }
    }
}

```

Answer: $\Omega(1)$ if the if statement is true, $O(n)$ if the if statement is false

```

6. void function6(int n) {
    int count = 0;
    for (int i = 1; i <= n/2; i++) {
        for (int j = 1; j <= n/3; j++) {
            for (int k = 1; k <= n/4; k++) {
                count++;
            }
        }
    }
    cout << count;
}

```

Answer: $\theta((n/12)^3)$

```

7. void function7(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j += i) {
            cout << "*";
        }
    }
}

```

Answer: $\theta(n * \log(n))$

```

8. void function8(int n) {
    int i = 1, s = 1;
    while (s <= n) {
        i++;
        s += i;
        cout << "*";
    }
}

```

Answer: $\theta(\sqrt{n})$

9. Processing Arrays

- a. Suppose you have an unsorted array of integers of length n and want to sum all the elements inside it. What is the running time of your algorithm? $\theta(n)$

- b. Suppose you have an unsorted array of integers of length n and want to determine if all the values inside are positive. What is the running time of your algorithm? $\theta(n)$
- c. Suppose you have a sorted array of integers of length n and want to determine the median value. What is the running time of your algorithm? $\theta(1)$

10. **T/F** $f(n) = 5n^2 + 4n + 8 \in \theta(n^2)$

If true, prove it by giving *integral* values for the required constants c_1 , c_2 , and n_0 . Choose the tightest values possible for the c_1 and c_2 constants. If false, show the contradiction.

$$5n^2 \rightarrow \Omega(n^2)$$

$$5n^2 + 4n + 8 < 5n^2 + 5n \text{ when } (n > 8), \text{ so } 5n^2 + 5n \rightarrow O(n^2)$$

Since $5n^2 < 5n^2 + 4n + 8 < 5n^2 + 5n$ becomes $\Omega(n^2) < f(n) = 5n^2 + 4n + 8 < O(n^2)$ then $f(n) = 5n^2 + 4n + 8 \in \theta(n^2)$ for this to be true.

The constants in this case would be $c_1 = 5$, $c_2 = 2$, $n_0 = 6$.